

Marine mammal bycatch estimation for Alaska fisheries

Brian Brost
Alaska Ecosystems Program
Marine Mammal Lab
Alaska Fisheries Science Center
NOAA Fisheries

Model-based estimation

For a given marine mammal and fishery:

$$y_{st} \sim \text{Binom}(N_{st}, p_{st})$$

$$N_{st} \sim \text{Pois}(\lambda_{st})$$

$$\log(\lambda_{st}) = \mathbf{x}'_s \boldsymbol{\beta} + \alpha_t$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma_{\beta}^2 \mathbf{I})$$

$$\alpha_t \sim \mathcal{N}(0, \sigma_{\alpha}^2)$$

$$\sigma_{\alpha}^2 \sim \text{IG}(r, q)$$

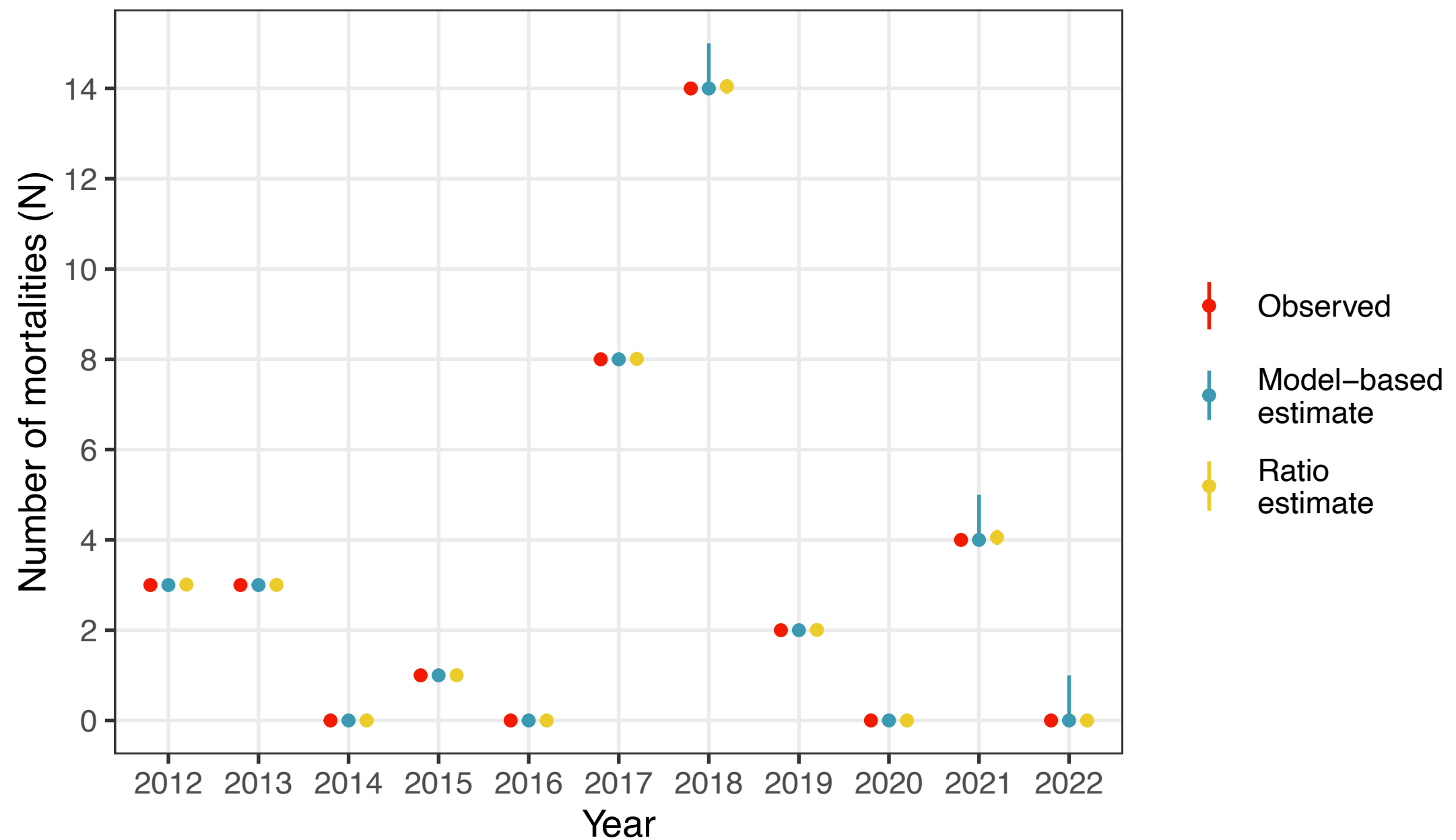
Strata (s) $\left\{ \begin{array}{l} \text{Week: } [1, 18), [18, 36), [36, 52) \\ \text{Vessel length: } [0, 60), [60, 125), 125+ \\ \text{Area: Bering Sea, Aleutian Islands, Gulf of Alaska} \end{array} \right.$

Year (t): 2012–2022

Annual bycatch

Ringed seal: BSAI flatfish trawl

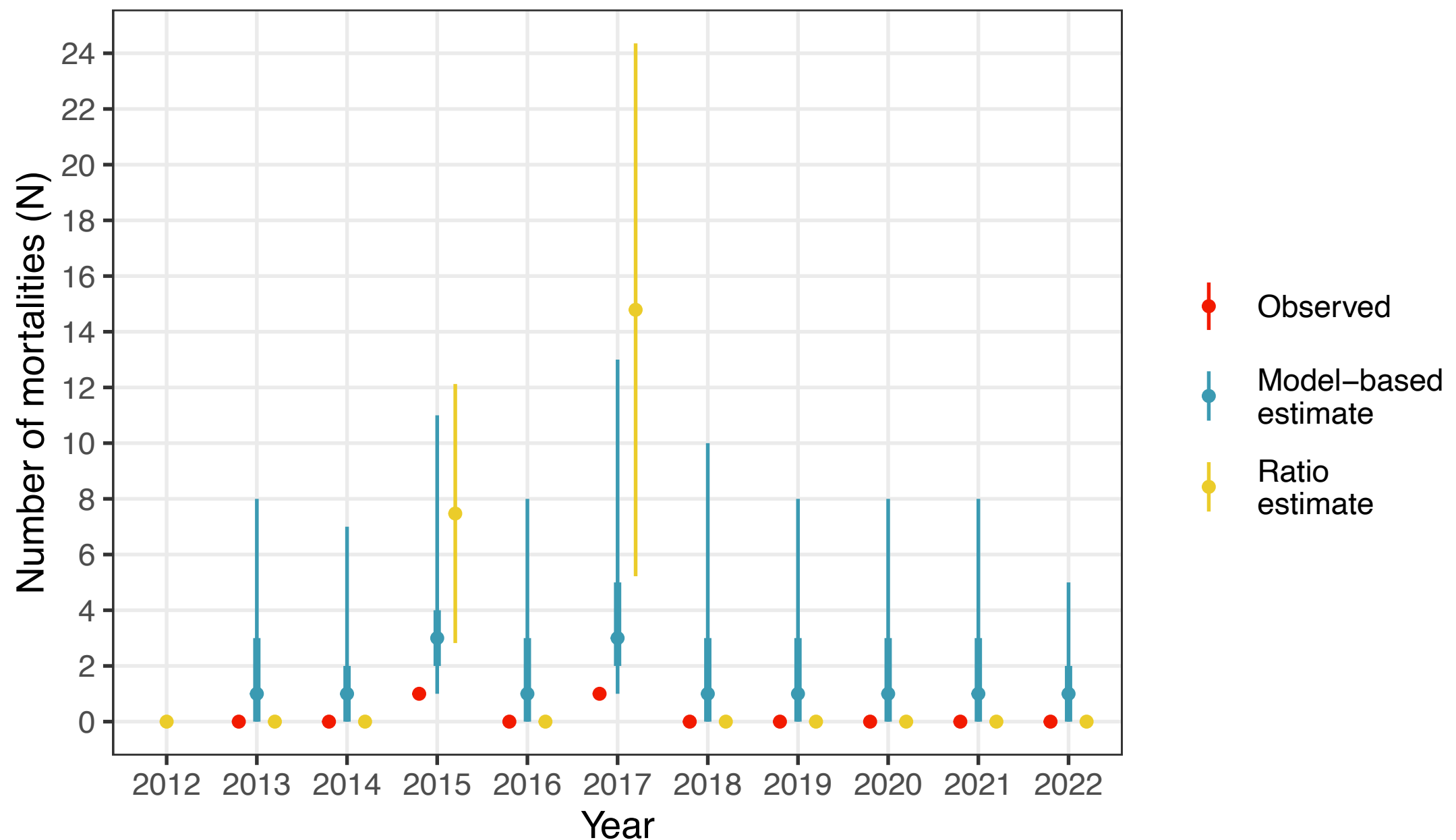
Observed and estimated mortalities



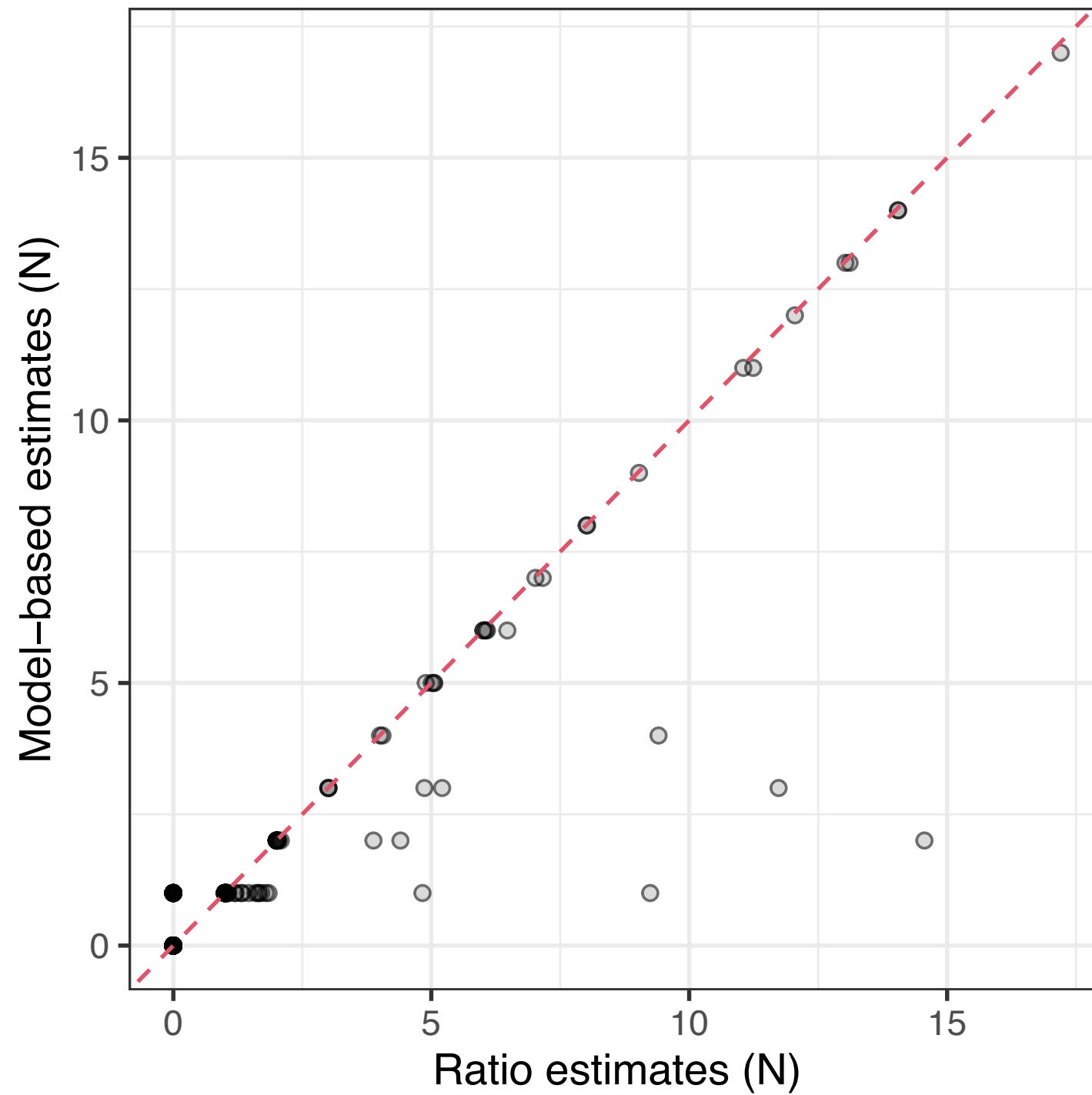
Annual bycatch

Steller sea lion / Eastern: GOA sablefish longline

Observed and estimated mortalities



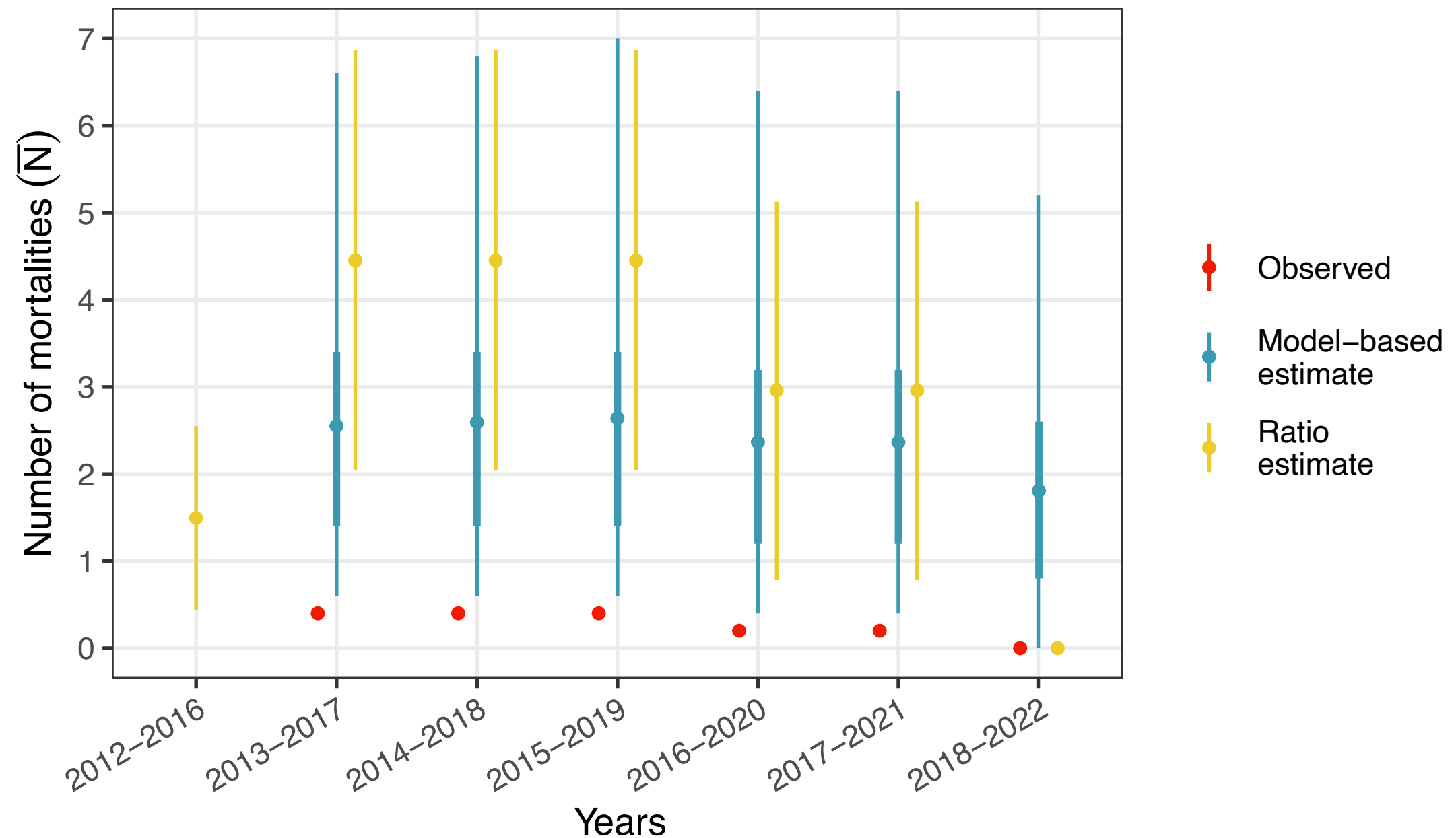
Model-based vs. ratio estimates



5-year averages

Steller sea lion / Eastern: GOA sablefish longline

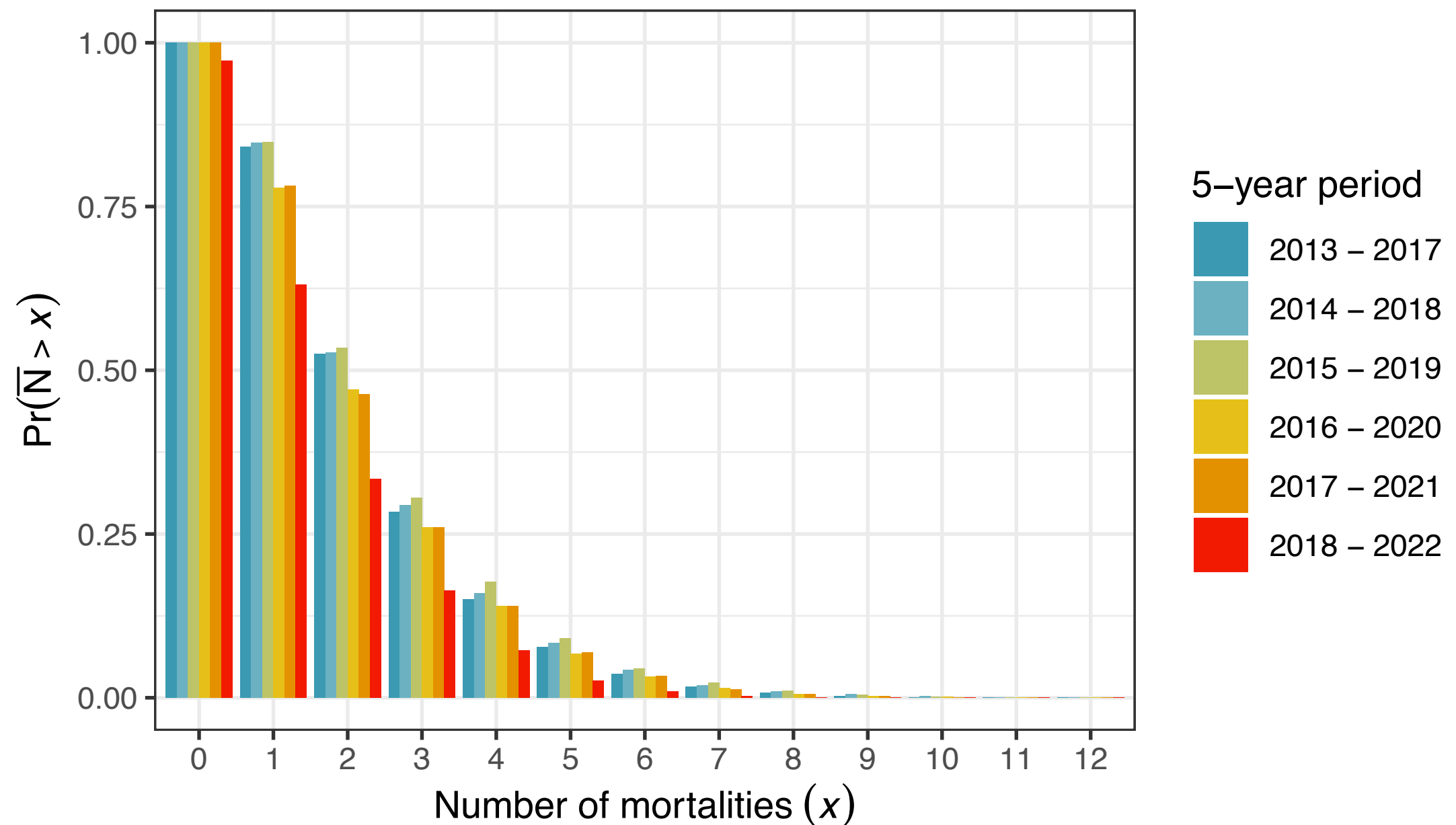
Average observed and estimated mortalities



5-year averages

Steller sea lion / Eastern: GOA sablefish longline

Probability of exceeding x mortalities



To do...

- Check model adequacy
- Convergence / mixing
- Incorporate additional bycatch events

